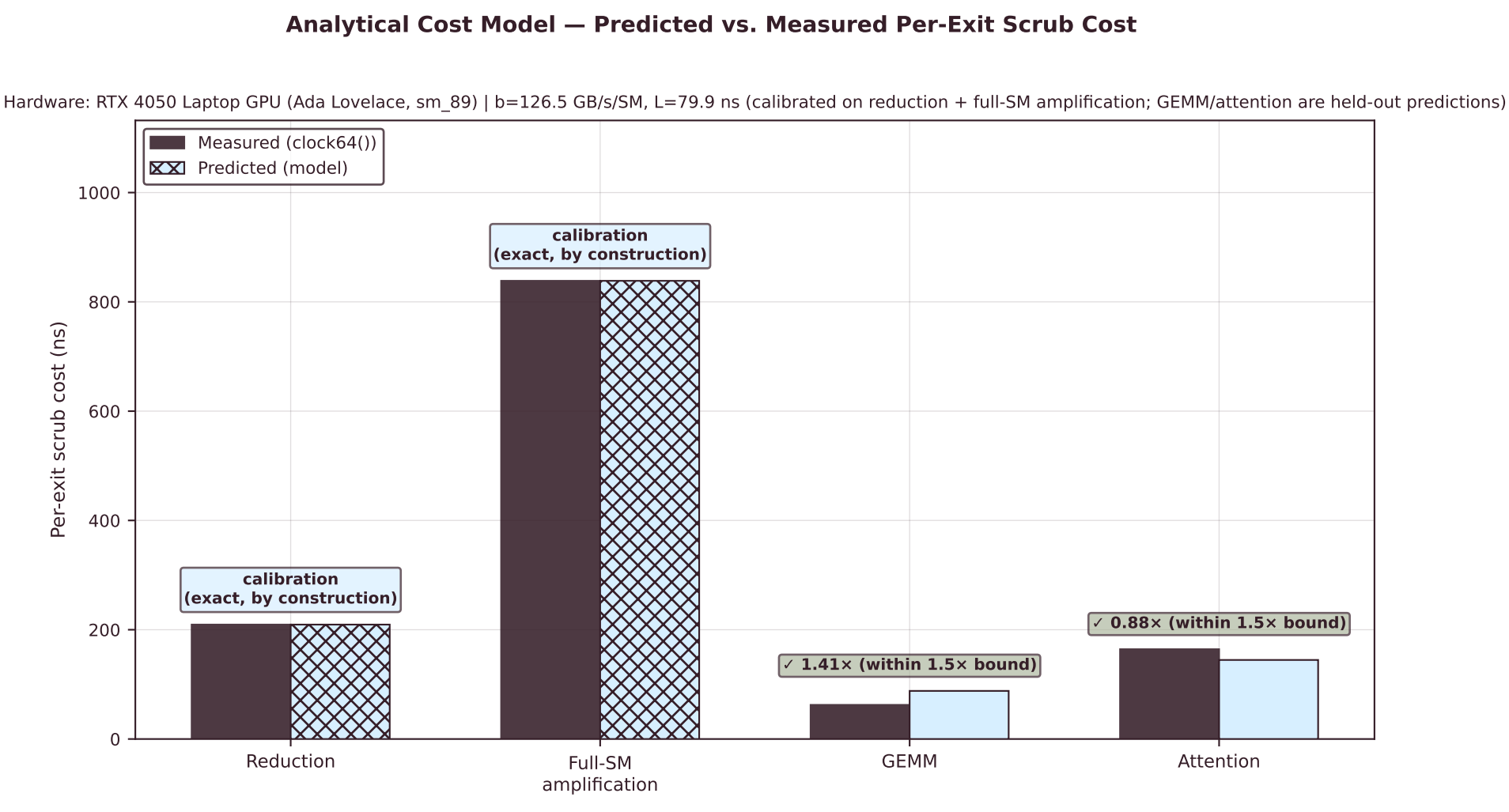
Analytical Cost Model — Predicted vs. Measured Per-Exit Scrub Cost
Hardware: RTX 4050 Laptop GPU (Ada Lovelace, sm_89) | b=126.5 GB/s/SM, L=79.9 ns (calibrated on reduction + full-SM amplification; GEMM/attention are held-out predictions)
Measured (clock64())
Predicted (model)
Per-exit scrub cost (ns)
0
200
400
600
800
1000
calibration (exact, by construction)
calibration (exact, by construction)
✓ 1.41× (within 1.5× bound)
✓ 0.88× (within 1.5× bound)
Reduction
Full-SM amplification
GEMM
Attention